California Housing Price Prediction

Machine Learning Project Report | Linear Regression

1. Project Objective

The objective of this project is to demonstrate a reproducible end-to-end machine learning workflow by building a Linear Regression model to estimate California house values. The work covers data loading, exploratory data analysis (EDA), data-quality checks, preprocessing assessment, model training, evaluation and reporting.

The project uses Python, Pandas, Matplotlib, Seaborn and Scikit-learn. The primary model is ordinary Linear Regression, as specified in the task.

2. Dataset Overview

The California Housing dataset is loaded through **sklearn.datasets.fetch_california_housing()**. It contains 20,640 observations, eight numerical input features and one numerical target. The target, named **Price** in the notebook, is expressed in units of $100,000; for example, a target value of 2.0 represents approximately $200,000.

Item	Description
Observations	20,640
Input features	8 numerical features
Target	Median house value (Price)
Feature examples	Median income, house age, average rooms, population, latitude and longitude
Train/test split	80% training (16,512 rows); 20% testing (4,128 rows)

3. Exploratory Data Analysis

The dataset was inspected using the first few records, its dimensions, column names, data types and descriptive statistics. Histograms were used to examine the distribution of house values, a scatter plot explored the relationship between median income (MedInc) and house value, and a correlation heatmap summarized pairwise linear relationships among numerical variables.

- **Price distribution:** the target distribution was examined to understand its spread and concentration.
- **Median income vs. price:** the scatter plot was used to inspect the relationship between household income and house value.
- **Correlation:** the heatmap provided a compact view of linear associations; correlation does not establish causation.

Dataset limitation: the target values are capped at approximately 5.0 (about $500,000). This ceiling can limit the model's ability to distinguish more expensive properties.

4. Data Quality and Preprocessing

Data-quality checks found no missing values and no duplicate records in the loaded dataset. All eight input features are numerical, so categorical encoding is not needed.

Check	Finding	Action
Missing values	0 in all columns	No imputation required
Duplicate rows	0	No duplicate removal required
Feature types	All numerical	No categorical encoding required
Feature scales	Different ranges	Retained original scales for ordinary Linear Regression

Feature scale assessment

The input features have different numerical ranges. For example, MedInc ranges from approximately 0.50 to 15.00, while Population ranges from 3 to 35,682. Standardization was considered but not applied. Ordinary least-squares Linear Regression does not require standardized features to generate predictions, unlike scale-sensitive methods such as K-nearest neighbors or support vector machines. The original feature values were therefore retained for this baseline model.

The target and input features were separated into **y** and **X**, respectively. The data was then divided into training and testing subsets using an 80:20 split and **random_state=42** to make the split reproducible.

5. Model: Linear Regression

Linear Regression models the target as a weighted combination of the input features plus an intercept. During training, Scikit-learn's **LinearRegression** estimator fits the coefficients by minimizing the sum of squared differences between observed and predicted target values.

The model was fitted using only the training data. Predictions were then generated for the held-out test data, which provides an estimate of performance on unseen observations.

6. Evaluation Metrics

Metric	Purpose	Interpretation
MAE	Mean absolute error	Average absolute prediction error
MSE	Mean squared error	Penalizes larger errors more strongly
RMSE	Square root of MSE	Error magnitude in the target's original units
R^2	Coefficient of determination	Explained variation relative to a mean baseline

Because the target is measured in $100,000 units, MAE and RMSE can be converted to approximate dollar amounts by multiplying by $100,000. MSE is in squared target units. R^2 is not a percentage accuracy score; it describes the proportion of target variation explained by the model relative to predicting the test-set mean.

7. Results and Interpretation

Evaluation metric	Result
Mean Absolute Error (MAE)	0.5332 (approximately $53,320)
Mean Squared Error (MSE)	0.5559
Root Mean Squared Error (RMSE)	0.7456 (approximately $74,558)
Training R^2	0.6126
Testing R^2	0.5758

The test R^2 of 0.5758 indicates that the model explains approximately 57.58% of the variation in house values in the test set, relative to a baseline that predicts the test-set mean. The MAE corresponds to an average absolute error of about $53,320, and the RMSE is about $74,558.

The training R^2 (0.6126) is moderately higher than the testing R^2 (0.5758), a difference of approximately 0.0368. This suggests a modest generalization gap. The model captures some meaningful structure in the data, but its predictions still have substantial error.

Diagnostic Analysis

- **Actual vs. predicted plot:** compares predicted values with observed test values. Predictions close to the ideal diagonal indicate closer agreement.
- **Residual analysis:** residuals are calculated as actual minus predicted values. A random scatter around zero is desirable; visible patterns may indicate model limitations.

8. Limitations

- The linear model may not capture nonlinear relationships or complex interactions between features.
- The dataset target is capped near $500,000, limiting evaluation at the high end of the market.
- The model's R^2 and error metrics indicate that a considerable portion of variation remains unexplained.
- The results depend on the selected train-test split; cross-validation would provide a more robust estimate.

9. Possible Improvements

- **Cross-validation:** evaluate the model across multiple folds to obtain a more stable performance estimate.
- **Regularization:** compare Ridge and Lasso Regression to assess whether coefficient constraints improve generalization.
- **Feature engineering:** investigate useful transformations and interactions while avoiding information leakage.
- **Nonlinear models:** compare tree-based regressors, such as Decision Tree or Random Forest, as optional extensions.
- **Further diagnostics:** inspect residual patterns and evaluate performance across different price ranges.

10. Conclusion

This project demonstrates the main stages of a machine learning workflow using the California Housing dataset and Scikit-learn. The Linear Regression baseline achieved a test R^2 of 0.5758, with an MAE of approximately $53,320. The model provides a useful starting point and a reproducible baseline for further experimentation. Future work can focus on cross-validation, feature engineering, regularized regression and nonlinear models to investigate potential improvements.

Note: This report summarizes the notebook results supplied for the project. Include the notebook's generated plots alongside this report if the submission portal permits separate visual attachments.